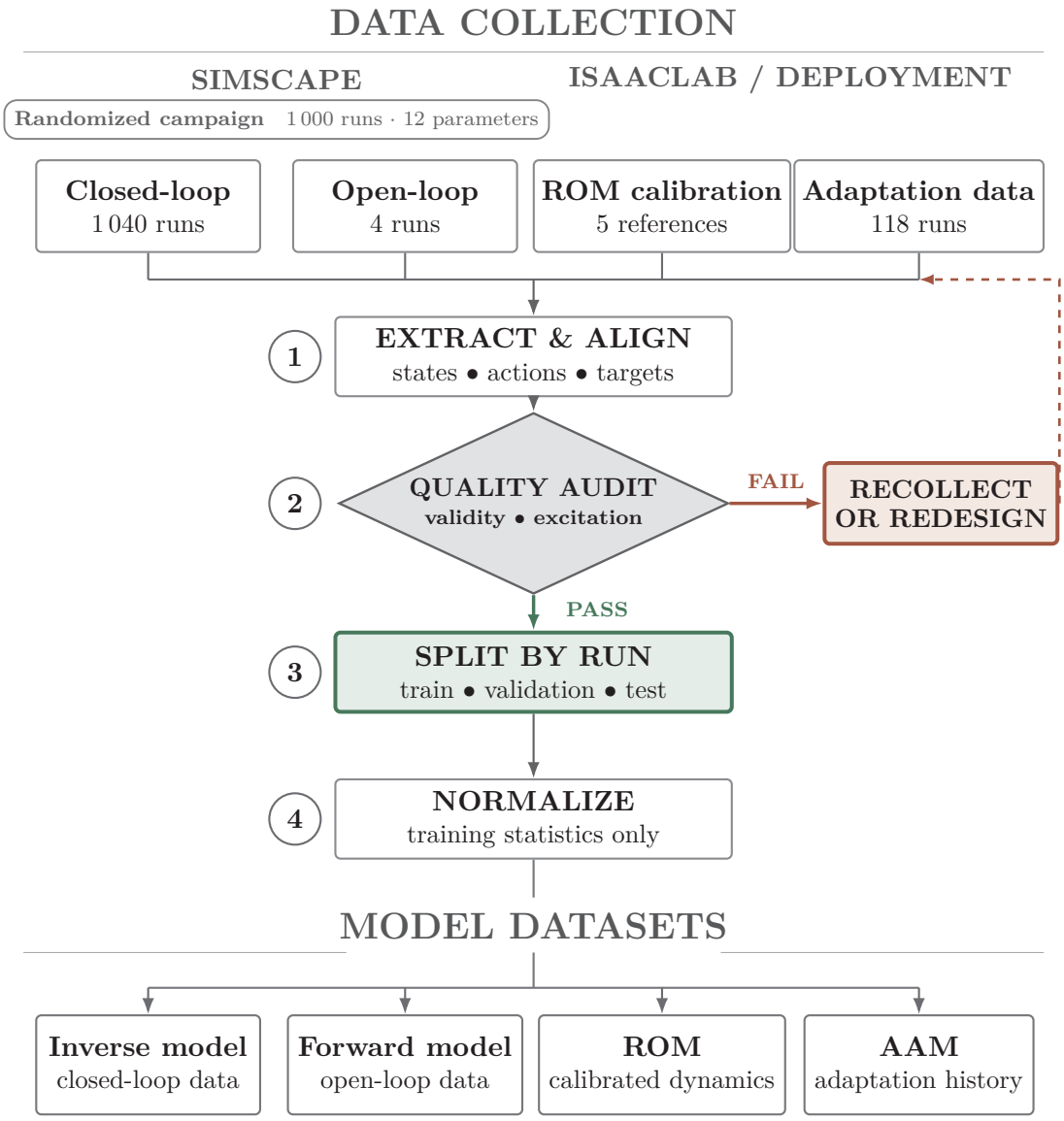


Figure 4.1: Data-preparation workflow. Two Simscape collections and two IsaacLab collections pass through common extraction, audit, run-level splitting and training-statistics normalization stages. The randomized Simscape campaign is provenance metadata rather than a pipeline stage. A failed audit returns the experiment to collection; accepted material forms the inverse, forward, ROM and AAM datasets.

4.1 Dataset Objectives and Design Requirements

Four requirements governed collection and preparation.

R1: realistic operating distribution. The learned models are deployed inside a tracking controller, so they must be accurate in the states that controller visits: coordinated multi-axis motion, gravity-loaded poses, moderate speeds. Data collected far from that distribution would be accurate where it does not matter.

R2: causal identifiability of the valve. A forward model is only useful to a policy if it responds to the command. This requires that the valve signal in the training data